

SMART INDUSTRIAL LIQUID MIXER

A PROJECT REPORT

Submitted by

JITHINRAJ K (KKE22AE026)
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to

the APJ Abdul Kalam Technological University
In partial fulfillment for the award of the Degree

of

Bachelor of Technology

in

Applied Electronics & Instrumentation



**Department of
Applied Electronics & Instrumentation**

Government Engineering College Kozhikode
Kannur Road, Westhill Post Office, Calicut
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MAY 2026

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DECLARATION

I undersigned hereby declare that the project report "SMART INDUSTRIAL LIQUID MIXER" , submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by we under supervision of Sumi George. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

*This is to certify that the project entitled ‘SMART INDUSTRIAL LIQUID MIXER’ submitted by **JITHINRAJ K (KKE22AE026), JUGAL DEV M (KKE22AE027), KEERTHANA V (KKE22AE028), and MINHAL RAHMAN (KKE22AE030)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in **Applied Electronics & Instrumentation** is a bonafide record of the project work carried out by them under my supervision and guidance. This report in any form has not been submitted to any other University or Institute for any purpose.*

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Dr. Baburaj M.

Head of the Department

ABSTRACT

Industrial liquid mixing is a fundamental operation in sectors ranging from pharmaceuticals and chemicals to food and beverage manufacturing. Traditional mixing processes are often hampered by inconsistencies from manual operation, the rigidity of conventional automation, and a lack of real-time process insights. This project addresses these challenges by designing and validating a Smart Industrial Liquid Mixer, envisioned as an adaptable, intelligent, and interconnected platform based on Industry 4.0 principles. The project employs a dual-validation methodology, beginning with high-fidelity computational modeling in LabVIEW to verify advanced control laws, including PWM-based regulation and automated batching sequences. These theoretical models were then translated into a functional hardware implementation featuring a microcontroller-based actuation system and a high-precision gravimetric feedback loop. By integrating a load-cell sensing unit with an optimized, user-friendly IoT dashboard, the system achieves precise ingredient ratios and provides real-time remote monitoring capabilities. This software interface transforms the mixer into a transparent, data-driven system, allowing operators to configure recipes and monitor dosing accuracy through a web-based environment. The final implementation successfully reconciles theoretical control simulations with practical hardware constraints, demonstrating a scalable, cost-effective solution for precision automation and digital transformation in modern industrial mixing environments.

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Chapter 1

INTRODUCTION

This chapter introduces the context, motivation, and scope of the Smart Industrial Liquid Mixer project. It establishes the critical need for advanced automation in industrial liquid mixing and details the transition from theoretical modeling to a functional, IoT-enabled hardware implementation. Finally, it outlines the specific problem addressed, its overall significance, and the boundaries of the current scope.

1.1 BACKGROUND AND MOTIVATION

Industrial liquid mixing is a fundamental operation across various sectors, including chemical, pharmaceutical, and food processing, where achieving precise consistency and quality is critical [1, 2]. Historically, these processes have been governed by manual intervention or conventional automation relying on Programmable Logic Controllers (PLCs), which have served as the backbone of modern manufacturing [3, 4]. However, traditional automation often lacks the flexibility required for rapid batch reconfiguration and seamless digital integration [5]. With the emergence of Industry 4.0, there is an increasing demand for systems that converge physical hardware with digital intelligence [6]. This project builds on foundational research by utilizing a dual-validation approach—first establishing system dynamics through computational modeling and then realizing these models through a functional, cost-effective hardware prototype featuring an optimized, software-defined user interface.

1.2 PROBLEM STATEMENT AND PROJECT SIGNIFICANCE

The primary problem addressed is the operational rigidity and high cost of current proprietary industrial mixing solutions. While PLC and SCADA-based systems provide reliability [7], they often lack native IoT connectivity and user-friendly, high-fidelity monitoring tools [8]. Furthermore, manual mixing methods lead to significant inconsistencies and material wastage in small-to-medium scale applications. The significance of this project lies in successfully bridging the gap between high-level simulations and physical implementation using accessible hardware [9, 10]. By integrating

a high-precision gravimetric feedback loop with an optimized, user-friendly IoT dashboard, the solution provides a transparent, data-driven framework that lowers the barrier to entry for intelligent automation in batch manufacturing.

1.3 SCOPE OF THE PROJECT

The scope of this project encompasses the complete development cycle of the Smart Industrial Liquid Mixer, treated as a single, integrated engineering effort. This work includes the following key areas:

- **System Modeling:** Utilizing LabVIEW to verify complex control logic, including PID-based regulation and automated batching sequences.
- **Hardware Implementation:** Developing a functional prototype featuring a microcontroller-based actuation system and a load-cell sensing unit for precise weight-based dosing.
- **Software Integration:** Designing an optimized web-based dashboard to expose the software side of the system, enabling real-time monitoring and remote recipe configuration.
- **Performance Validation:** Proving the system's accuracy and stability by reconciling theoretical control simulations with practical hardware performance.

Chapter 2

LITERATURE SURVEY

This chapter presents a comprehensive review of existing industrial control paradigms and current research trends in automation technology. It first examines the established role of traditional PLC systems in batch processing. The survey then shifts focus to the growing feasibility of utilizing microcontrollers for hardware-in-the-loop implementation and IIoT convergence. Finally, this review identifies the current research gap and establishes the unique novelty of the proposed integrated mixer platform.

2.1 TRADITIONAL CONTROL ARCHITECTURES IN BATCH PROCESSING

The PLC remains the industry benchmark for deterministic, sequential control in liquid mixing and filling operations [1, 3]. These systems are typically integrated with SCADA systems to manage complex sequences, such as ingredient addition and agitation cycles [2, 5]. While these architectures provide high reliability and are widely used in chemical and pharmaceutical sectors [8], they are often characterized by high proprietary costs and a lack of native flexibility for small-scale, customizable applications. This establishes a clear need for more accessible, software-defined control environments that can emulate PLC logic on open-source hardware.

2.2 HARDWARE REALIZATION AND IIOT CONVERGENCE

Recent advancements in automation have seen a significant shift toward utilizing high-performance microcontrollers to bridge the gap between physical processes and digital intelligence [6, 10]. Modern research highlights that microcontrollers, when integrated with robust actuation drivers, can effectively manage industrial-grade tasks such as real-time pump control and sensor data acquisition [4]. Furthermore, the convergence of control systems with the IIoT allows for enhanced operational transparency [8]. By moving away from rigid hardware buttons toward optimized web-based dashboards, operators can achieve a higher degree of remote accessibility and real-time process visualization, which is central to the Industry 4.0 transition.

2.3 SENSING STRATEGY: GRAVIMETRIC VS. VOLUMETRIC PRECISION

Industrial mixing requires high dosing accuracy to maintain product quality, especially in sensitive applications like fertilizer or chemical production [7, 9]. While volumetric methods are common, recent literature emphasizes the superior accuracy of gravimetric or weight-based feedback loops. Utilizing high-resolution sensing units allows the controller to eliminate errors caused by liquid viscosity or temperature fluctuations [9]. The integration of these feedback loops into a unified software environment enables the implementation of advanced control laws, such as PWM-based dosing, which ensures that final ingredient ratios remain within strict industrial tolerances.

2.4 RESEARCH GAP AND PROJECT NOVELTY

A synthesis of current literature confirms that while simulation tools and hardware controllers exist independently, there is a lack of work demonstrating a cost-effective, unified platform that consolidates core industrial control, high-precision gravimetric feedback, and an optimized IoT user interface [4, 6]. This project’s novelty lies in successfully reconciling theoretical models developed in MATLAB and LabVIEW with a functional hardware prototype. By exposing the software side through an optimized dashboard, the system provides a transparent, data-driven framework that proves the feasibility of high-fidelity industrial automation on a scalable, low-cost platform.

2.5 CONTRIBUTION OF KEY LITERATURE

The following literature provided foundational knowledge, design precedents, and technical validation for the Smart Industrial Liquid Mixer project:

[1] F. A. A. Mohamed and S. B. R. Ramli, “Automated mixing process system by using programmable logic controller and arduino uno” (2023): This paper is directly relevant as it explores an automated mixing system that utilizes both a PLC and an Arduino Uno. It justifies the project’s core methodology: that a low-cost microcontroller can handle industrial-grade automation tasks while maintaining acceptable accuracy. By integrating these platforms, the paper highlights the strengths

of microcontrollers in terms of cost-effectiveness and flexibility, solidifying the choice of the ESP32 for core functionality in this project.[1]

[2] K. Vadivukkarasi and S. Bhargava, “Mixing and filling operations using plc, scada and iot” (2024): This paper is crucial as it successfully integrates a PLC with an IoT module (ESP8266EX) for mixing and filling. The methodology directly informs the system architecture of the proposed project, validating the use of a Wi-Fi-enabled microcontroller like the ESP32 for remote control. It demonstrates the practical application of IIoT concepts, specifically addressing the remote monitoring functionalities essential for the project’s web-based dashboard.[2]

[3] S. Ahmad and V. K. Shukla, “Industrial automation through plc and scada” (2025): This study provides a comprehensive overview of the fundamental architectures of traditional automation. It establishes the ”gold standard” benchmarks for robust control and supervisory management. This helps define the reliability requirements for the proposed mixer, ensuring that the new low-cost design meets the functional standards typically expected from expensive industrial-grade hardware.[3]

[4] V. S. S. Mungi, “Automated batch process system using tia openness software” (2025): This paper is instrumental in defining the necessary components for a generic automated batch process. The work provides foundational knowledge on discrete manufacturing steps, which is directly applicable to the liquid mixing sequence design. It explores the potential of incorporating advanced software-defined automation to optimize sequential systems, guiding the decision toward the structured operational flow used in this project.[4]

[5] T. A. Khan, “Harnessing artificial intelligence for optimum performance in industrial automation” (2025): This study provides a practical perspective on achieving optimum performance through smart, data-driven approaches. It validates the goal of overcoming the limitations of fixed-logic control systems. The research emphasizes performance enhancements derived from intelligent integration, supporting the project’s foundation on modern, connected technology rather than isolated hardware.[5]

[6] A. K. Mbeah and E. Normanyo, “Automation of a pharmaceutical batch mixer using plc and scada” (2024): This research serves as a direct case study for pharmaceutical-grade mixing. It outlines critical control sequences, including precise ingredient addition and agitation control. The paper’s documentation of simulation-based validation allows this project to benchmark its low-cost design against proven

industrial efficacy, ensuring the gravimetric feedback loop is modeled with high precision.[6]

[7] S. Jegan et al., “PLC based automatic mixing system for chemical industry” (2023): This work describes a mixing system specifically for the chemical industry, focusing on accurate raw material proportioning. This relates directly to the precision needs of the smart liquid mixer. The system’s procedural model for mixing and quality testing offers a strong framework for developing the project’s sequential control logic and validating the accuracy of the weight-based dosing system.[7]

[8] N. Dhameliya, “Revolutionizing plc systems with ai: A new era of industrial automation” (2023): This paper investigates the effects of combining control hardware with higher-level intelligence. The discussion on optimizing process control informs the design of the project’s future-ready features. It positions the proposed system as a forward-looking design, acknowledging the trend of integrating data analytics with core control infrastructure to maximize operational efficiency.[8]

[9] M. L. Nesabagyam et al., “Smart automation of fertilizer ratio mixer using plc” (2023): This work provides a technical foundation by detailing the implementation of a ratio mixer for precise ingredient proportioning. The study’s focus on maintaining specific component ratios offers a direct technical blueprint for the liquid ratio aspect of the current project. It reinforces the necessity of precise sensing mechanisms for achieving high accuracy in resource-intensive batch processes.[9]

Chapter 3

OBJECTIVES

This chapter details the foundational goals and the structured engineering approach adopted for the development of the Smart Industrial Liquid Mixer. By establishing clear primary and technical objectives, the project ensures a systematic transition from high-fidelity computational modeling to a robust, physical realization that meets modern industrial standards.

3.1 PRIMARY OBJECTIVES

The overarching goal of this project is to engineer a unified automation platform that bridges the gap between conventional industrial control and modern digital transparency. While PLCs remain the industry gold standard for deterministic and rugged process control [3], this project seeks to augment that reliability with the flexibility of open-source, IoT-enabled hardware. The primary objectives are:

- **Hybrid Automation Development:** To design a control system that replicates the sequential reliability of PLC-based batching while utilizing a high-performance microcontroller for enhanced computational flexibility.
- **Precision Gravimetric Regulation:** To implement a closed-loop dosing system that utilizes weight-based feedback, ensuring high-accuracy ingredient proportioning regardless of environmental variables like fluid temperature or viscosity.
- **Industry 4.0 Integration:** To develop a full-stack software interface that exposes internal system variables to a web-based dashboard, enabling remote monitoring, digital calibration, and data-driven operational transparency.

3.2 TECHNICAL OBJECTIVES

To fulfill the primary objectives, the following technical milestones have been defined. These tasks ensure that every hardware component and software module is validated against theoretical benchmarks.

Table 3.1: Technical objectives

Sl. No.	Technical Objective	Description and Target)
1	System modeling and identification	To perform mathematical modeling in LabVIEW to determine the transfer of the mixing process and tune PID constants for stable regulation
2	Firmware architecture design	To develop a multi-tasking firmware on the ESP32 platform that handles real-time sensor interrupts, PWM actuation, and Wi-Fi stack management simultaneously
3	High-precision sensing integration	To interface a load cell with a 24-bit ADC(HX711) to achieve subgram accuracy in the mixing ratio, replacing traditional time-based or volumetric flow estimation
4	IoT dashboard development	To build a responsive, user-friendly web-dashboard that serves as a software side exposure for the operator, featuring live telemetry and recipe controls
5	Hardware-loop-in-the-loop validation	To verify the physical prototype's response against the initial simulations, quantifying the dosing error percentage and system settling time under various batch conditions

3.3 PROJECT METHODOLOGY

The project methodology is divided into three comprehensive stages, ensuring that the final smart implementation is rooted in solid engineering theory:

- **Stage I- Analytical Modeling and Simulation:** The initial phase involves utilizing LabView for system identification. A virtual model of the liquid mixer is created to simulate various disturbance factors. Control strategies, including PWM and sequential batching logic, are tested within this environment to

establish the performance ceiling of the system.

- **Stage II- Hardware Prototyping and Control Integration:** Based on the validated models, the physical system is assembled. This involves the integration of a microcontroller-based control unit with 5V submersible pumps and L298N motor drivers. The core of the hardware is the gravimetric feedback loop, which uses a high-resolution load cell to provide real-time mass data to the controller, allowing for high-precision ingredient dosing that mimics the accuracy of specialized industrial mixers [10].
- **Stage III- Software-Side Exposure and IoT Deployment:** The final stage focuses on the smart capabilities of the system. An optimized, user-friendly web dashboard is developed and hosted on the ESP32. This interface allows the user to interact with the software parameters such as setting the target weight for different liquids and provides a visual representation of the process through live charts. This integration ensures that the machine is not a black box but a transparent, interconnected industrial asset.

3.4 SUPERVISORY INTERFACE AND SYSTEM OPERABILITY

To improve system usability and operational flexibility, a web-based supervisory interface was integrated with the Smart Liquid Mixer system. The interface enables wireless interaction with the ESP32 controller for monitoring and controlling the liquid dosing process in real time. Unlike conventional industrial HMI systems, the proposed interface utilizes lightweight web technologies, making the prototype compact, cost-effective, and suitable for embedded IoT applications.

- **Web Dashboard Interface:** A browser-based dashboard was developed using HTML, CSS, and JavaScript to provide an interactive user interface for the system. The dashboard allows users to start or stop the process, monitor live weight values, and observe the operational status of pumps and mixing stages.
- **Recipe Selection and Process Configuration:** The system supports recipe-based operation, allowing predefined liquid quantity combinations to be selected and executed automatically. This improves process repeatability and simplifies user interaction during operation.

- **Live Process Monitoring:** Real-time communication between the ESP32 and the web dashboard enables continuous monitoring of the dosing process, including weight measurement updates and dispensing progress.
- **Controlled Liquid Dispensing:** PWM-assisted pump control was incorporated to reduce overshoot and improve dispensing accuracy near the target weight values, resulting in smoother and more efficient operation.

Chapter 4

PROPOSED METHOD

The proposed methodology for the Smart Industrial Liquid Mixer is developed as a compact gravimetric liquid dosing and process automation platform intended for prototype-scale industrial applications. The system combines embedded control, real-time mass measurement, adaptive actuation, and wireless supervisory monitoring to achieve precise multi-liquid dispensing and automated mixing operations. Unlike conventional volumetric approaches that depend on fixed timing or estimated flow rates, the proposed system adopts a gravimetric control methodology in which dispensing accuracy is continuously validated using real-time load measurements.

The development approach followed a simulation-assisted prototyping methodology. During the initial phase of development, LabVIEW was utilized as a pre-prototyping environment for analyzing process sequencing, validating state-machine behavior, visualizing operational flow, and studying dosing logic under different conditions. The LabVIEW environment was not used for direct hardware interfacing in the final implementation. Instead, the validated operational strategy was translated into embedded firmware running on the ESP32 microcontroller. The final hardware prototype operates independently using wireless communication and a lightweight browser-based dashboard developed using HTML, CSS, and JavaScript technologies.

The proposed architecture emphasizes modularity, adaptive process control, low-cost implementation, and IoT-based operational transparency. The system integrates gravimetric feedback sensing, PWM-assisted flow regulation, recipe-driven sequencing, and real-time telemetry visualization into a unified embedded automation framework.

4.1 SYSTEM OVERVIEW AND CORE FUNCTIONALITY

The Smart Industrial Liquid Mixer is designed as an embedded gravimetric batching and mixing platform capable of executing sequential liquid dispensing operations according to predefined recipe parameters. The primary objective of the system is to automate ingredient dosing while maintaining accurate liquid ratios through continuous mass-based feedback acquisition.

The operational core of the system is centered around an ESP32 microcontroller

that supervises sensor acquisition, actuator control, process sequencing, wireless communication, and dashboard hosting. The controller continuously monitors the cumulative vessel mass using a load cell measurement subsystem and dynamically regulates the liquid dispensing process according to the programmed setpoints.

Unlike conventional timer-based dosing systems, the proposed system adopts a closed-loop gravimetric feedback architecture. Real-time weight data obtained from the load cell is continuously compared against the target recipe values to determine pump operation status and flow regulation behavior. This feedback-oriented approach improves dosing repeatability and minimizes dependency on fluid characteristics such as viscosity or flow variation.

The system also incorporates adaptive PWM-assisted flow control to reduce overshoot during the final stage of liquid dispensing. High-speed filling is used during the initial phase of dosing, while gradual deceleration is introduced near the target value to improve process precision. Once both ingredients are dispensed, a relay-controlled mixing stage is activated to ensure homogeneous blending of the final mixture.

The ESP32 additionally hosts a browser-based supervisory dashboard that allows wireless process interaction through Wi-Fi connectivity. The dashboard provides recipe selection capability, live telemetry visualization, tare operation, process initiation, and system monitoring through an intuitive user interface.

4.2 FUNCTIONAL SYSTEM BLOCKS AND INTERCONNECTIONS

The proposed architecture is organized into multiple functional subsystems to improve implementation flexibility, process reliability, and scalability. Each subsystem performs a dedicated operational role within the overall automation framework.

4.2.1 Control and Communication Unit

The ESP32 microcontroller functions as the computational and supervisory core of the entire system. It is responsible for executing the embedded process logic, controlling actuator operation, generating PWM signals, processing gravimetric feedback data, and maintaining wireless dashboard communication.

The controller operates using a sequential state-machine-based logic structure in which each operational phase is executed according to process completion conditions.

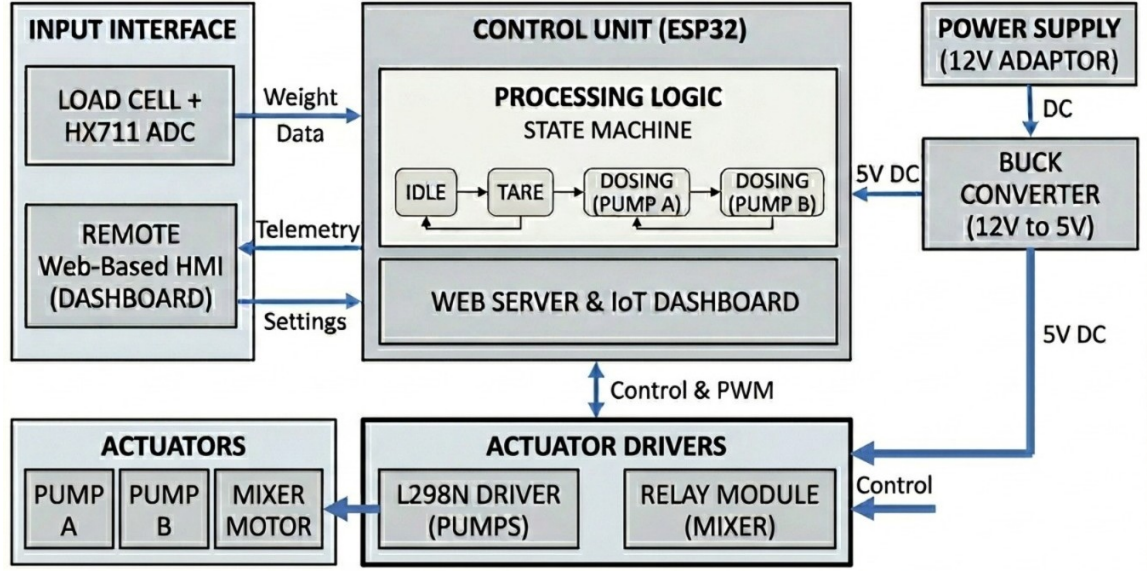


Figure 4.1: Functional block diagram of the Smart Industrial Liquid Mixer

The ESP32 continuously acquires digital weight measurements from the HX711 interface module and determines the corresponding actuator response based on recipe parameters and control conditions.

Integrated Wi-Fi functionality within the ESP32 enables the hosting of a lightweight embedded web server. This allows the system to provide remote dashboard accessibility without requiring dedicated industrial SCADA infrastructure or external communication gateways. The communication subsystem supports real-time telemetry exchange between the hardware prototype and the browser-based supervisory interface.

4.2.2 Gravimetric Sensing and Measurement Subsystem

The sensing subsystem is based on a load cell integrated with the HX711 24-bit analog-to-digital conversion module. This subsystem forms the primary feedback mechanism for gravimetric dosing control.

The load cell continuously measures the cumulative mass of the dispensing vessel during operation. Mechanical deformation generated by applied load is converted into an electrical signal, which is amplified and digitized by the HX711 module before being processed by the ESP32.

By utilizing gravimetric feedback instead of fixed-time dispensing, the system

achieves improved dosing consistency and reduced sensitivity to pump flow variation. The measurement subsystem also supports digital tare calibration, allowing the system to establish an accurate zero reference before each operational cycle.

The high-resolution sensing capability of the HX711 module improves process stability and allows the controller to perform adaptive flow regulation during the final stage of dosing.

4.2.3 Actuator Drives and Liquid Dispensing Mechanism

The actuator subsystem consists of two miniature DC liquid pumps and a mixer motor integrated through dedicated driver stages. The pumps are controlled using an L298N motor driver module, while the mixer motor is operated through a relay-based switching mechanism.

The L298N driver provides directional control and PWM-assisted speed regulation for both pumps. During operation, the pumps initially function at higher duty cycles to achieve rapid liquid transfer. As the measured mass approaches the target setpoint, the PWM duty cycle is gradually reduced to minimize overshoot and improve gravimetric accuracy.

This adaptive flow reduction strategy enables smoother dosing behavior compared to conventional binary ON/OFF control systems. Sequential dispensing logic ensures that the second ingredient is introduced only after successful completion of the first dosing stage.

The relay-controlled mixer motor is activated after completion of both dosing operations to provide agitation and homogeneous mixing of the final liquid composition.

4.2.4 Power Supply and Voltage Regulation

The system utilizes a 12V DC adapter as the primary energy source for the prototype. A buck converter module is integrated into the power distribution stage to generate regulated 5V DC required for the embedded controller, sensing subsystem, pumps, and relay circuitry.

The regulated power architecture improves electrical stability and reduces noise interference during sensitive gravimetric measurements. Proper common-ground integration between sensing and actuation subsystems ensures reliable signal acquisition and stable actuator performance during operation.

The power subsystem is designed with scalability considerations, allowing future integration of additional actuators, communication modules, or higher-capacity process equipment without significant architectural redesign.

4.3 SEQUENTIAL CONTROL STRATEGY AND OPERATIONAL FLOW

The Smart Industrial Liquid Mixer follows a deterministic sequential control strategy based on gravimetric process validation and embedded state-machine execution. The operational logic was initially analyzed in the LabVIEW simulation environment to study sequencing behavior, dosing transitions, process timing, and control flow visualization before hardware realization.

The final implementation executes the validated operational sequence directly through ESP32 firmware. The control methodology prioritizes gravimetric accuracy, adaptive flow regulation, and reliable stage-wise process execution.

4.3.1 Flowchart Explanation

The operational sequence of the system proceeds through the following stages:

1. **System Initialization:** The ESP32 initializes the gravimetric sensing subsystem, establishes wireless communication, and activates the embedded web dashboard server.
2. **Recipe Configuration:** The operator selects predefined liquid quantity parameters through the browser-based dashboard interface.
3. **Digital Tare Calibration:** The system performs gravimetric zero calibration to eliminate vessel weight and establish an accurate baseline reference.
4. **Primary Liquid Dispensing:** Pump A is activated, and the controller continuously compares live mass readings against the first recipe setpoint.
5. **Adaptive Flow Regulation:** As the measured mass approaches the target quantity, PWM duty cycle is progressively reduced to achieve smoother flow deceleration and reduced overshoot.

6. **Secondary Liquid Dispensing:** Upon completion of the first dosing stage, Pump B is activated for dispensing the second ingredient according to the predefined recipe value.
7. **Agitation and Mixing:** After successful completion of both dosing phases, the relay module activates the mixer motor for homogeneous blending of the liquid composition.
8. **Process Completion and Telemetry Update:** All actuators are deactivated, final process data is updated on the dashboard, and the system returns to the idle operational state.

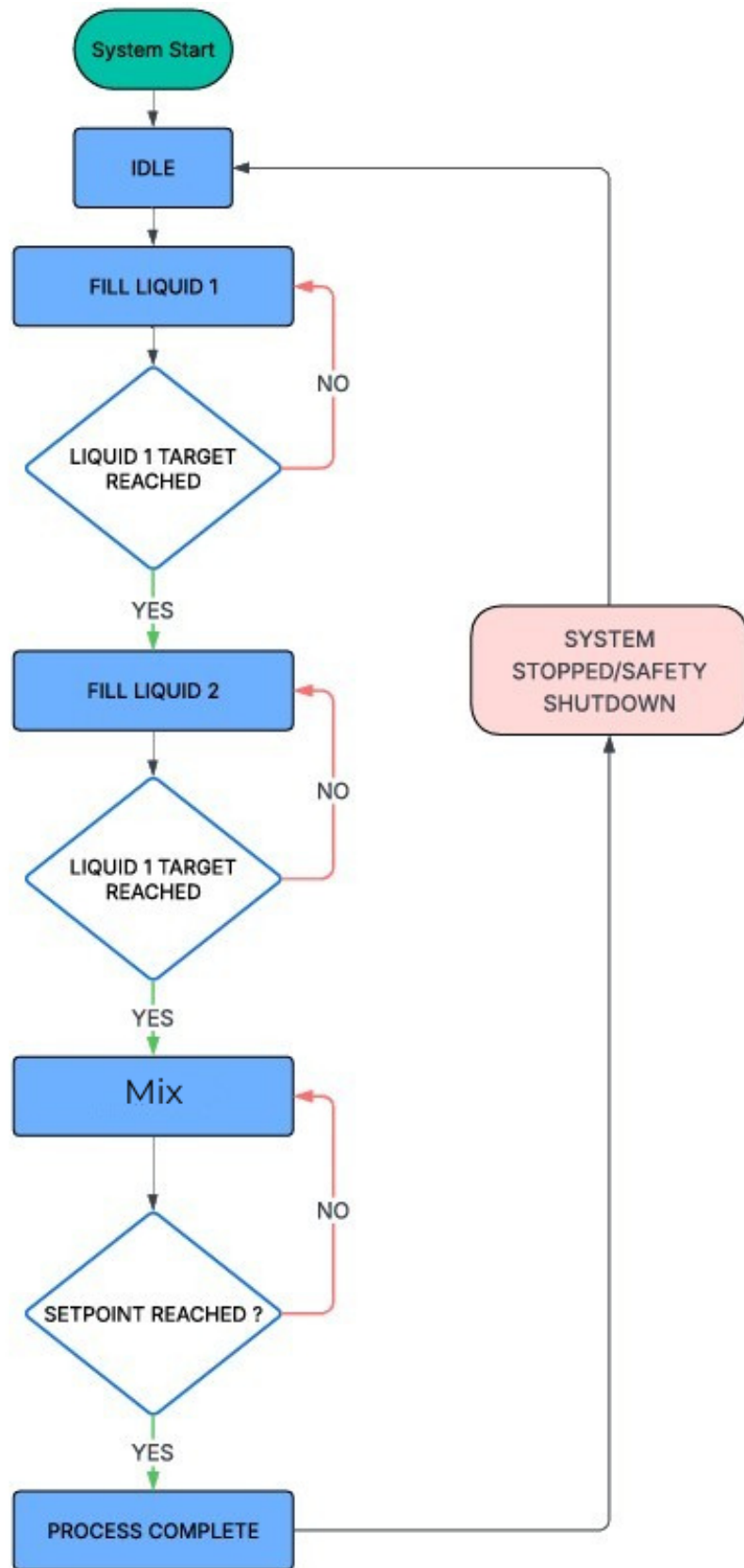


Figure 4.2: Sequential control flow of the Smart Industrial Liquid Mixer

4.3.2 Implementation Considerations

- **Modular Embedded Architecture:** The subsystem-oriented design allows future integration of additional sensors, actuators, or communication modules without extensive firmware restructuring.
- **Adaptive Gravimetric Control:** PWM-assisted flow regulation improves dosing precision by minimizing transient overshoot during the terminal phase of liquid dispensing.
- **Wireless Supervisory Monitoring:** The embedded web dashboard provides lightweight IoT-based supervisory interaction without requiring dedicated SCADA infrastructure.
- **Prototype Scalability:** The proposed architecture can be extended for industrial batching, cloud telemetry integration, remote diagnostics, and advanced process analytics applications.

4.3.3 Summary of the Proposed System

The proposed Smart Industrial Liquid Mixer demonstrates the practical integration of gravimetric sensing, embedded automation, adaptive actuation, and IoT-enabled supervisory monitoring within a compact process-control platform. By utilizing real-time mass feedback, PWM-assisted dosing control, and recipe-driven sequential operation, the system achieves reliable and flexible liquid dispensing suitable for prototype-scale industrial applications. The modular ESP32-based architecture and lightweight web-dashboard interface further improve operational transparency, scalability, and future extensibility for intelligent industrial automation systems.

Chapter 5

SYSTEM DESIGN AND IMPLEMENTATION

This chapter provides the detailed technical blueprint and engineering realization of the proposed Smart Industrial Liquid Mixer. It defines the system architecture, component interfaces, and hardware implementation required to achieve automated liquid dispensing and mixing. The design adopts a modular and layered approach, integrating sensing, control, and actuation subsystems while ensuring stable power distribution and reliable real-time operation. Emphasis is placed on practical implementation, accurate measurement, and coordinated control of all system components. The following sections describe the hardware architecture, component specifications, and pin configurations of the ESP32-based control unit and its associated modules.

5.1 SYSTEM DESIGN DEFINITION

System design involves translating the conceptual control requirements into a structured hardware architecture that defines components, interfaces, and data pathways. The Smart Industrial Liquid Mixer is designed as a modular automation system with the ESP32 microcontroller acting as the central control hub, coordinating sensing, processing, and actuation tasks. The design emphasizes reliable sensor-actuator interaction, simplified wiring, and stable power regulation to ensure consistent performance under practical conditions.

The system adopts a weight-based control approach using a load cell interfaced with the HX711 amplifier to measure liquid quantity accurately. The ESP32 processes this sensor data and executes sequential control logic to operate pumps and mixer based on predefined conditions. The design ensures that each subsystem operates within defined electrical and functional limits, maintaining robustness against noise and fluctuations. This structured implementation provides a dependable foundation for automated operation while allowing future enhancements and scalability.

5.2 DETAILED HARDWARE ARCHITECTURE: THE CIRCUIT BLUEPRINT

The hardware architecture represents the physical realization of the control and automation logic, integrating power, control, sensing, and actuation subsystems into a cohesive system. The design follows a layered approach, separating low-voltage control circuits from high-power actuation components to ensure safe and reliable operation. The ESP32 microcontroller serves as the central processing unit, interfacing with all peripheral modules and executing control logic based on sensor feedback and user inputs.

The sensing subsystem consists of a load cell connected to the HX711 amplifier, which converts analog strain signals into digital data for accurate weight measurement. The actuation subsystem includes two DC pumps controlled through an L298N motor driver, enabling controlled liquid transfer, and a mixer motor operated using a relay module for safe switching. The power architecture utilizes a 12V DC supply for high-power components, while a buck converter provides regulated 5V output for the ESP32 and sensor modules. Proper grounding, wiring practices, and component placement are maintained to reduce electrical noise and ensure stable operation.

The circuit diagram illustrates the complete interconnection of all system components, including ESP32 pin mapping, motor driver inputs, relay connections, and sensor interfacing. It serves as the primary reference for hardware implementation and system integration.

5.2.1 Subsystem Overview

The system is organized into functional subsystems, each designed to perform a specific role in the automated liquid mixing process while operating in coordination under the control of the ESP32 microcontroller. This modular structure simplifies implementation, improves reliability, and enables easier troubleshooting.

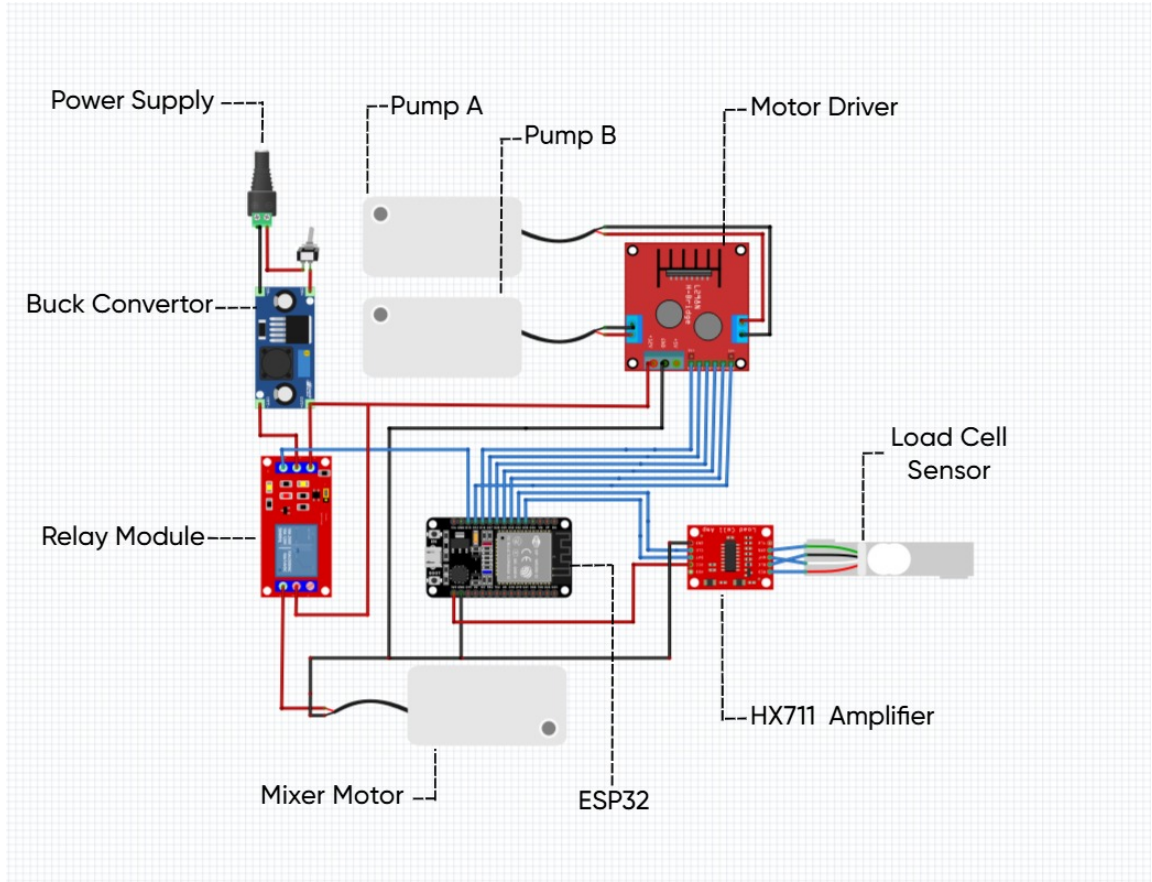


Figure 5.3: Detailed system circuit diagram of the Smart Industrial Liquid Mixer

Control Subsystem: Built around the ESP32 microcontroller, this subsystem acts as the central processing unit of the system. It is responsible for executing control algorithms, reading sensor data, and generating control signals for actuators. The ESP32 continuously monitors system inputs and coordinates the sequence of operations for pumps and mixer, ensuring proper timing and synchronization.

Sensing Subsystem: The sensing unit comprises the load cell and HX711 amplifier module. The load cell detects weight variations caused by liquid accumulation, while the HX711 converts these analog signals into high-resolution digital data. This subsystem provides real-time feedback to the controller, enabling accurate and repeatable liquid measurement. Proper calibration and stable mounting of the load cell are critical for consistent performance.

Actuation Subsystem: The actuation subsystem includes two DC pumps and a mixer motor responsible for liquid transfer and mixing. The pumps are controlled

using the L298N motor driver, which supplies the required current and enables controlled operation. The mixer motor is operated through a relay module, providing safe switching and isolation from control circuitry. The ESP32 determines the sequence and duration of operation for each actuator based on system logic.

Power Management Subsystem: This subsystem ensures stable and regulated power distribution across all components. A 12V DC supply powers the pumps and mixer motor, while a buck converter steps down the voltage to 5V for the ESP32, HX711 module, and relay circuit. Proper grounding and separation of power and control circuits minimize noise and ensure reliable operation of sensitive components.

Together, these subsystems form an integrated system architecture that enables accurate measurement, controlled actuation, and reliable operation of the Smart Industrial Liquid Mixer.

5.3 COMPONENT SPECIFICATION AND PIN CONFIGURATION

The selection of components for the Smart Industrial Liquid Mixer is based on functionality, availability, and compatibility with embedded control systems. The ESP32 microcontroller serves as the central processing unit, interfacing with sensing and actuation modules while enabling wireless communication for user interaction. Peripheral components are chosen to ensure accurate measurement, reliable actuation, and stable operation under varying conditions.

The system integrates sensing elements for feedback, driver circuits for actuation, and regulated power modules to maintain consistent voltage levels. Component interfacing is designed to minimize signal distortion, ensure electrical safety, and support real-time processing requirements. The detailed functional mapping of ESP32 pins with connected hardware modules is provided in Table 5.2.

5.3.1 Precision Sensing Modules

The sensing subsystem of the Smart Industrial Liquid Mixer is designed to provide accurate and reliable measurement of liquid quantity using a strain-gauge-based load cell integrated with the HX711 precision amplifier. The load cell operates on the principle of resistance variation caused by mechanical deformation when a load is applied, converting the applied weight into a proportional electrical signal. This signal is typically of very low magnitude and requires amplification before it can be

Table 5.2: ESP32 microcontroller pin configuration and functional mapping

Function	ESP32 Pin	Rationale / Connection Type
HX711 Data (DT)	GPIO 21	Receives digital weight data from HX711 module
HX711 Clock (SCK)	GPIO 22	Provides clock for HX711 communication
Pump A Control (IN1)	GPIO 26	Controls direction/input for Pump A through L298N
Pump A Control (IN2)	GPIO 27	Secondary control input for Pump A
Pump B Control (IN3)	GPIO 32	Controls direction/input for Pump B through L298N
Pump B Control (IN4)	GPIO 33	Secondary control input for Pump B
PWM Enable (ENA)	GPIO 25	PWM speed control for Pump A
PWM Enable (ENB)	GPIO 14	PWM speed control for Pump B
Relay Control Signal	GPIO 18	Controls relay module for mixer motor switching
Logic Supply	3.3V	Powers ESP32 and HX711 module
Pump / Relay Supply	5V (Buck Converter)	Powers pumps and relay module
Input Supply	12V Adapter	Primary supply input for the overall system
Common Ground	GND	Shared reference ground for all modules and power stages

processed by the microcontroller.

The HX711 module performs both amplification and analog-to-digital conversion, producing a high-resolution digital output corresponding to the measured weight. This allows the ESP32 microcontroller to continuously monitor the amount of liquid being dispensed in real time. To ensure consistent performance, software-based signal conditioning techniques such as averaging and filtering are employed to minimize fluctuations caused by electrical noise and environmental disturbances.

The accuracy of the sensing subsystem is highly dependent on proper mechanical configuration, including stable mounting of the load cell and uniform distribution of applied load. Any misalignment, vibration, or uneven force application can introduce measurement errors, and therefore structural support and careful calibration are essential. By combining precise signal amplification with real-time processing, the sensing module enables reliable weight-based control, improving the overall efficiency

and repeatability of the liquid mixing process.

5.3.2 Actuation and Power Switching Interfaces

The actuation subsystem is responsible for executing physical operations such as liquid transfer and mixing. Two DC pumps are employed to dispense liquids sequentially, and their operation is controlled through the L298N motor driver module. The driver provides sufficient current amplification and isolates the microcontroller from high-power loads, ensuring safe operation. Directional control and switching logic are managed through digital signals from the ESP32, enabling precise timing and sequencing of pump operation.

The mixing process is carried out using a DC motor controlled via a relay module. The relay acts as an electromechanical switch, allowing the ESP32 to control higher voltage circuits without direct electrical contact. This isolation enhances system safety and prevents damage to low-voltage control circuitry.

To improve reliability, the actuation logic incorporates safeguards such as controlled startup, defined operating intervals, and immediate shutdown capability through an emergency stop mechanism. These features ensure safe handling of actuators and prevent continuous operation in fault conditions. The coordinated interaction between pumps and mixer ensures proper blending of liquids according to system requirements.

5.3.3 Power Regulation and Isolation

The system employs a structured multi-level power architecture to ensure stable and efficient operation of both control and actuation components. A 12V DC adapter is used as the primary power source, selected to accommodate potential future expansion involving higher power pumps and actuators. In the current implementation, this input voltage is stepped down to 5V using a buck converter, which supplies power to the mini submersible pump through the motor driver. The ESP32 microcontroller and HX711 amplifier module are powered using the regulated 3.3V output of the ESP32, ensuring compatibility and minimizing electrical noise in sensitive measurement circuits. Proper grounding is maintained across all modules to establish a common reference and reduce signal fluctuations, while high-current paths associated with motor operation are physically and electrically separated from low-power signal lines to limit electromagnetic interference. Additionally, a main power switch

is placed before the buck converter to control the overall system supply, enabling safe startup and shutdown while preventing unnecessary power consumption during idle conditions. This power regulation and isolation strategy enhances system reliability, protects components from voltage variations, and ensures consistent performance during operation.

5.4 DESIGN VALIDATION AND IMPLEMENTATION READINESS

The implemented system was validated through practical testing to evaluate the performance of sensing and actuation subsystems under real operating conditions. Initial tests were conducted using simple ON/OFF control of the pump, where the motor operated at full speed until the desired weight was reached. It was observed that due to fluid inertia and response delay in the system, the pump continued to dispense liquid even after receiving the stop command, resulting in overshoot and reduced accuracy.

To overcome this limitation, a two-stage control strategy was implemented using PWM control through the motor driver enable pin. In this approach, the pump operates at full speed during the initial phase of filling to ensure rapid liquid transfer. As the measured weight approaches the target value, the pump speed is reduced using PWM, enabling finer control and minimizing overshoot.

Experimental observations indicated that the two-stage control approach significantly improved dispensing accuracy and repeatability compared to simple ON/OFF control. The system demonstrated improved response near the target value, reduced error, and smoother operation. This enhancement highlights the effectiveness of combining sensor feedback with adaptive control techniques in embedded systems.

The overall system was further evaluated for stability, response time, and reliability. The ESP32 successfully processed sensor data and executed control logic in real time without noticeable delay. The system is therefore considered functionally complete and ready for demonstration, providing a practical foundation for further development and scalability.

Chapter 6

RESULTS

This chapter presents the results obtained from both simulation and practical implementation of the Smart Industrial Liquid Mixer. The system was initially analyzed using LabVIEW simulation to validate the control logic under ideal conditions. Subsequently, the system was implemented using hardware components and evaluated under real operating conditions. The results are analyzed to study system accuracy, response behavior, and the effect of different control strategies.

The performance of the system is examined using two control approaches: basic ON/OFF control and an improved two-stage PWM control method. A web-based dashboard is also integrated to provide real-time monitoring and control. The comparison between simulation and experimental results highlights the influence of real-world factors such as fluid inertia, sensor delay, and system dynamics on overall performance.

6.1 SIMULATION RESULTS USING LABVIEW

The system control logic was initially developed and verified using LabVIEW simulation. The simulation provided a graphical environment to visualize system behavior and validate the sequence of operations. The interface was designed to represent the real system, including input selection, control logic execution, and output response.

The simulation process begins with the user interface where input parameters are defined. The system then executes the selection logic, followed by the filling logic, where the simulated tank level increases based on control conditions. Finally, the system displays the output state after completion of the dosing process.

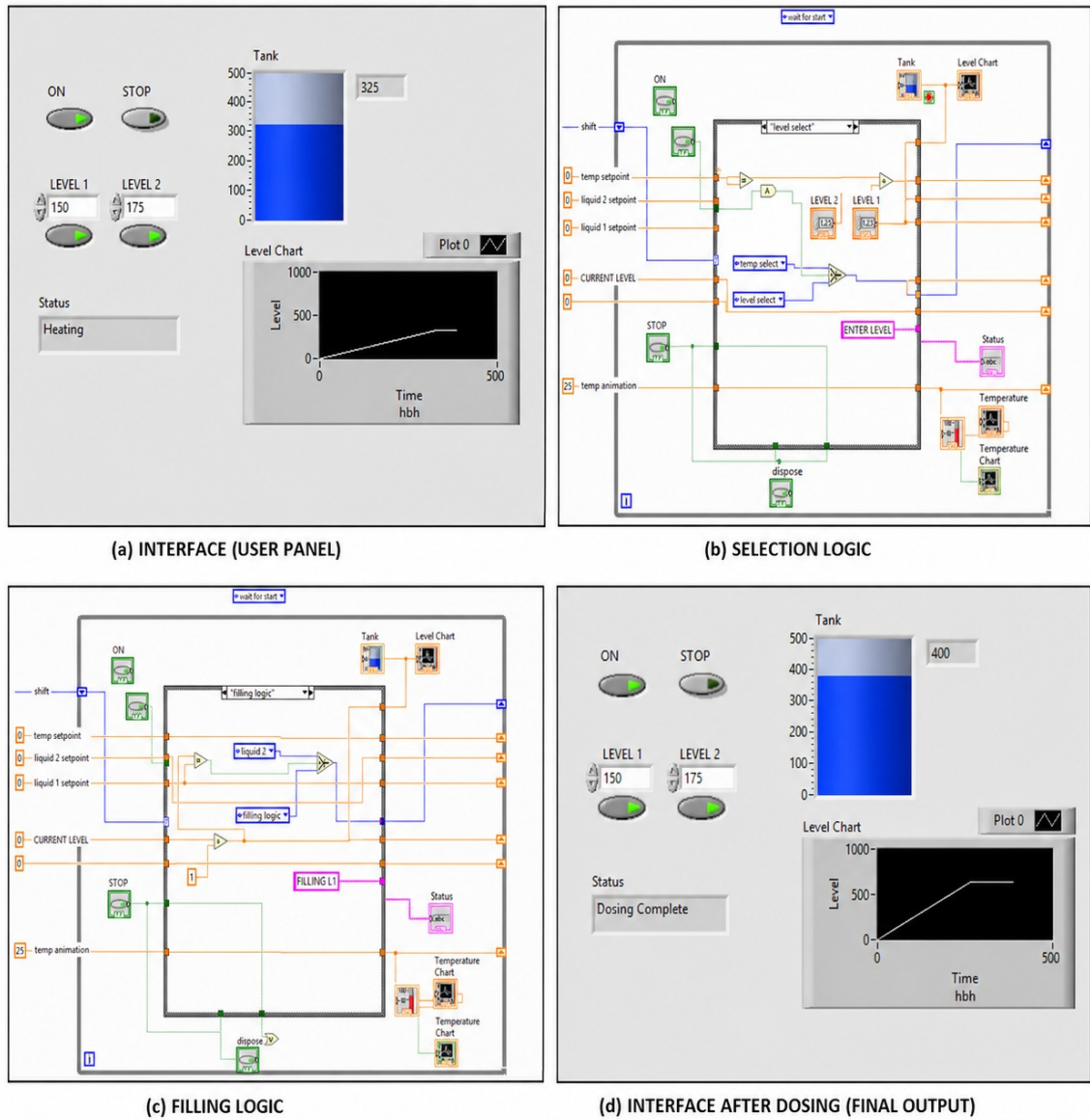


Figure 6.1: LabVIEW Simulation Stages

The simulation results show smooth and ideal system behavior, with the output reaching the target value without overshoot. Since the simulation environment does not include real-world effects such as delay, inertia, or noise, the system response appears highly accurate. These results confirm the correctness of the designed control logic and serve as a reference for evaluating practical performance.

6.2 WEB-BASED CONTROL AND MONITORING INTERFACE

A web-based dashboard was developed to enable real-time monitoring and control of the system. The dashboard communicates with the ESP32 microcontroller through a wireless network, allowing remote operation without the need for physical switches.

The interface supports recipe-based operation, where users can define, add, modify, and execute different liquid mixing configurations. Each recipe includes parameters such as quantity and process conditions, enabling flexible and customizable operation. The dashboard also provides real-time system status, including connection status, operational state, and actuator activity.

Live weight data from the load cell is displayed continuously, allowing accurate monitoring of the dispensing process. A graphical visualization is included to observe system behavior over time. Control options such as start, stop, and tare are provided for complete system interaction. An emergency stop function is also available, ensuring immediate shutdown during abnormal conditions and enhancing system safety.

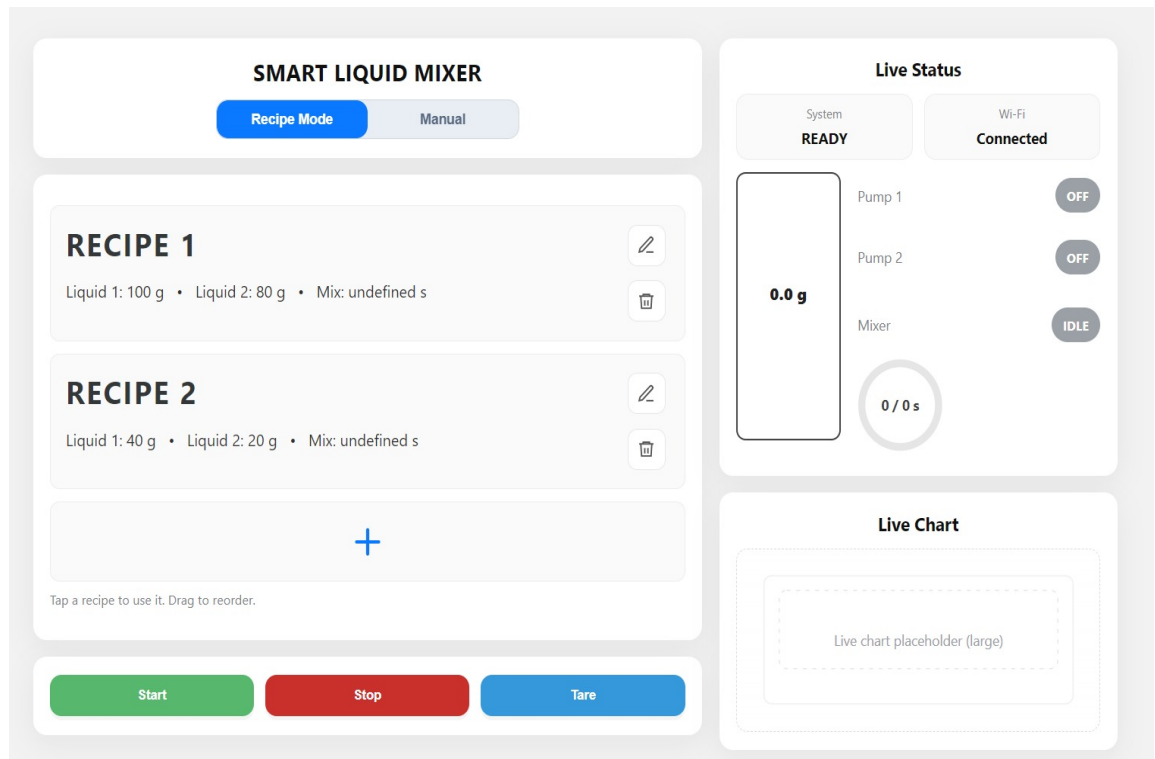


Figure 6.2: Web Dashboard Interface

The integration of the web dashboard improves system usability and demonstrates the application of IoT-based control in automation systems, enabling efficient and

user-friendly operation

6.3 HARDWARE PROTOTYPE IMPLEMENTATION

The developed system was implemented as a physical prototype integrating all hardware components, including the ESP32 microcontroller, load cell with HX711 amplifier, L298N motor driver, and mini submersible pump. The components were assembled and interconnected based on the designed circuit configuration, ensuring proper power distribution and signal interfacing.

The load cell was mounted securely to ensure stable weight measurement, and the pump was positioned to enable controlled liquid transfer into the container. The entire setup was powered through a regulated supply using a buck converter, ensuring safe operation of both control and actuation components.

The prototype was tested under practical conditions, and the system demonstrated successful integration of sensing, control, and actuation modules. The physical implementation validated the feasibility of the proposed design and provided a platform for experimental evaluation of system performance.



Figure 6.3: Hardware Prototype

6.4 EXPERIMENTAL RESULTS USING ON/OFF CONTROL

The system was initially tested using basic ON/OFF control, where the pump operates at full speed until the target weight is reached. This method represents a simple threshold-based control approach.

During experimentation, it was observed that the system exhibited moderate overshoot in the range of 10–15 gram. This occurs due to fluid inertia and delay between issuing the stop command and the actual cessation of liquid flow. Additionally, sensor response delay contributes to late detection of the target condition.

Table 6.1: ON/OFF Control Results

Trial	Target (g)	Actual (g)	Error (g)
1	200	212	+12
2	200	214	+14
3	200	211	+11
4	200	215	+15
5	200	213	+13

As a result, the system lacks precise control near the setpoint, leading to reduced accuracy and variation across trials. Although the method is simple, it is not suitable for applications requiring precise liquid dispensing.

6.5 EXPERIMENTAL RESULTS USING TWO-STAGE PWM CONTROL

To improve system accuracy, a two-stage control strategy was implemented using PWM. In this approach, the pump operates at full speed during the initial filling stage to ensure rapid liquid transfer. As the measured weight approaches the target value, the pump speed is reduced using PWM, allowing finer control of the dispensing process.

This method effectively compensates for fluid inertia and system delay by reducing flow rate near the setpoint. As a result, the system achieves better synchronization between sensor feedback and actuation.

Experimental results show a significant improvement in accuracy, with errors reduced to approximately 2–6 gram. The system exhibits smoother response and improved repeatability compared to ON/OFF control. The gradual transition between

high-speed and low-speed operation ensures both efficiency and precision, making the approach suitable for practical automation applications.

Table 6.2: Two-Stage PWM Control Results

Trial	Target (g)	Actual (g)	Error (g)
1	200	205	+5
2	200	203	+3
3	200	202	+2
4	200	206	+6
5	200	204	+4

6.6 COMPARATIVE ANALYSIS

A comparison between ON/OFF control and two-stage PWM control clearly demonstrates the improvement achieved in system performance. The ON/OFF method results in moderate overshoot due to abrupt stopping behavior, while the PWM-based method provides gradual control and significantly reduces error.

The two-stage control approach improves accuracy by approximately 65–70% compared to ON/OFF control. This highlights the importance of adaptive control strategies in handling real-world system dynamics such as delay and fluid inertia. The PWM method also enhances system stability and repeatability across multiple trials.

Table 6.3: Performance Comparison

Control Method	Average Error (g)	Performance
ON/OFF	13	Moderate accuracy
PWM (Two-Stage)	4	High accuracy

6.7 DISCUSSION

The comparison between simulation and experimental results reveals the gap between ideal and real-world system behavior. While simulation results showed perfect control without overshoot, practical implementation introduced factors such as delay, inertia, and sensor noise, which significantly affected performance.

The ON/OFF control method fails to account for these dynamic effects, resulting in reduced accuracy. In contrast, the two-stage PWM control successfully addresses

these limitations by introducing gradual control near the target value. The integration of sensing, control, and user interface through the web dashboard demonstrates a complete embedded automation system.

Overall, the results confirm that the proposed system is capable of performing controlled liquid dispensing with improved accuracy and reliability.

6.8 OBSERVATIONS

Key highlights of the final design include:

- Simulation results represent ideal system behavior without real-world disturbances
- Practical implementation introduced delay, inertia, and sensor noise
- ON/OFF control resulted in moderate overshoot (10 to 15 g)
- Two-stage PWM control significantly reduced error (2 to 6 g)
- System accuracy improved by approximately 65 to 70%
- Load cell performance improved after calibration
- Web dashboard enabled real-time monitoring and control

Chapter 7

CONCLUSION

This chapter presents the overall conclusion of the Smart Industrial Liquid Mixer project along with the practical limitations observed during implementation and the possible future enhancements of the system. The chapter summarizes the performance and achievements of the developed prototype and discusses the scope for further improvements in industrial automation and intelligent liquid dosing applications.

7.1 CONCLUSION

The developed Smart Industrial Liquid Mixer system successfully demonstrated an automated liquid dosing and mixing process using ESP32, load cell sensing, and web-based monitoring. The system was designed to perform accurate dispensing of two liquids based on predefined quantity values and execute controlled mixing operations through an integrated automation sequence. The implementation combined hardware and software components including the HX711 load cell module, L298N motor driver, relay-controlled mixer mechanism, and a responsive web dashboard interface. The prototype was capable of monitoring real-time weight values, controlling pump operation, and managing the overall dosing process through a wireless network interface.

The project initially focused on simulation and logical validation using LabVIEW-based analysis and control strategies. These simulated studies helped in understanding the behavior of the dosing process, flow control logic, and automated sequencing prior to hardware implementation. The practical prototype developed later validated the major functionalities proposed during the simulation stage. Experimental testing showed that the system was capable of performing automatic liquid dispensing with acceptable accuracy for a prototype-level implementation. The integration of PWM-based speed reduction near the target quantity improved the overall stability of the dosing process and helped reduce overshooting during liquid transfer operations.

The developed web dashboard enhanced the usability of the system by enabling recipe-based operation, manual quantity selection, system monitoring, and wireless interaction with the ESP32 controller. The live monitoring section provided real-time

visualization of pump status, system state, and measured weight values, making the system more interactive and user-friendly. The project also demonstrated the practical integration of embedded systems, sensor interfacing, motor control, and IoT-based communication within a single automation platform.

Overall, the Smart Industrial Liquid Mixer project successfully demonstrated the practical implementation of an automated liquid dosing and mixing system using embedded control, weight-based measurement, and IoT-enabled monitoring techniques. The project provided valuable understanding in automation system design, sensor interfacing, process control, and real-time industrial monitoring concepts relevant to Applied Electronics and Instrumentation engineering.

7.2 LIMITATION

Although the implemented prototype achieved the intended objectives, certain practical limitations were observed during testing. The system utilized small 5V prototype pumps, which introduced minor fluctuations in liquid flow characteristics. Variations in liquid inertia and residual dripping after pump shutdown occasionally resulted in small measurement deviations near the target values. The accuracy of the system also depended on proper load cell calibration and stable placement of the weighing platform.

Since the implementation was developed as a compact laboratory prototype, industrial-grade flow control components and precision valves were not incorporated in the current system. Environmental disturbances, vibrations, and uneven mounting conditions could also affect the accuracy of the load cell readings during operation. The current system mainly focused on prototype-level automation and therefore does not include industrial safety interlocks, advanced fault detection, or closed-loop industrial process controllers. The mixing mechanism was also implemented using a simple low-power DC motor arrangement suitable for prototype validation rather than heavy industrial mixing applications.

7.3 FUTURE SCOPE

The project can be further enhanced in several ways for future industrial applications. Higher precision can be achieved by integrating industrial dosing pumps, solenoid

valves, and advanced flow sensors. The control strategy can be improved using PID-based adaptive flow control for better dynamic response and reduced settling error. Additional liquid channels can also be incorporated to support multi-component industrial mixing operations.

Cloud-based monitoring, mobile application support, and remote data logging can further improve accessibility and system supervision. Future versions of the system may also include automatic viscosity compensation, AI-assisted recipe optimization, and industrial HMI integration for large-scale process automation environments. The implementation of industrial communication protocols such as Modbus, MQTT, or OPC-UA can also improve compatibility with modern industrial automation systems.

Further developments can include automatic cleaning systems, level monitoring for storage tanks, fault diagnosis mechanisms, and intelligent recipe management features. Integration with PLC-based control systems and industrial SCADA platforms can transform the prototype into a scalable industrial liquid processing solution suitable for chemical, food processing, pharmaceutical, and beverage manufacturing industries.

References

- [1] F. A. A. Mohamed and S. B. R. Ramli, “Automated mixing process system by using plc and arduino uno,” *Malaysian Journal of Industrial Technology (MJIT)*, vol. 7, no. 1, pp. 10–17, 2023.
- [2] K. Vadivukkarasi and S. Bhargava, “Mixing and filling operations using plc, scada and iot,” in *7th International Conference on Circuit Power and Computing Technologies (ICCPCT)*, 2024.
- [3] S. Ahmad and V. K. Shukla, “Industrial automation through plc and scada,” *International Journal for Research in Applied Science Engineering Technology (IJRASET)*, vol. 13, no. 7, 2025.
- [4] V. S. S. Mungi, “Automated batch process system using tia openness software,” in *International Conference in Advances in Power, Signal, and Information Technology (APSIT)*, 2025.
- [5] T. A. Khan, “Harnessing artificial intelligence for optimum performance in industrial automation,” in *Eng. Proc. 2024*, vol. 76, 2025, p. 105.
- [6] A. K. Mbeah and E. Normanyo, “Automation of a pharmaceutical batch mixer using plc and scada,” *Automation, Control and Intelligent Systems*, vol. 12, no. 4, pp. 71–107, 2024.
- [7] S. Jegan, B. Jagan, M. L. Nesabagyam, and N. K. Kumar, “Plc based automatic mixing system for chemical industry,” in *Third International Conference on Ubiquitous Computing and Intelligent Information Systems (ICUIS)*, 2023.
- [8] N. Dhameliya, “Revolutionizing plc systems with ai: A new era of industrial automation,” *American Digits: Journal of Computing and Digital Technologies*, vol. 1, no. 1, pp. 33–48, 2023.
- [9] M. L. N. Bagyam, S. Jegan, S. Rajasekar, T. Vembu, and S. Vishwabarathi, “Smart automation of fertilizer ratio mixer using plc,” in *International Conference on Energy, Materials and Communication Engineering (ICEMCE)*, 2023.

ANNEXURE A: MAPPING TO PROGRAM OUTCOMES

LIST OF PROGRAM OUTCOMES

- PO1: Engineering Knowledge
- PO2: Problem Analysis
- PO3: Design/Development of Solutions
- PO4: Conduct Investigations of Complex Problems
- PO5: Engineering Tool Usage
- PO6: Engineer and Society
- PO7: Ethics
- PO8: Individual and Team Work
- PO9: Communication
- PO10: Project Management and Finance
- PO11: Life-long Learning

MAPPING THE PROJECT TO PROGRAM OUTCOMES

PO	Correlation	Justification
PO1	3	Applied embedded systems, sensor interfacing, load cell measurement, PWM control, and IoT-based automation concepts to develop the automated liquid dosing system.
PO2	3	Analysed liquid dosing inaccuracies, overshooting effects, and automation requirements to develop an efficient control strategy for the system.
PO3	3	Designed and developed a compact automated liquid mixing system integrating ESP32, HX711, pumps, relay modules, and web-based monitoring.
PO4	2	Conducted experimental testing, calibration, simulation studies, and hardware validation to analyse system performance and dosing accuracy.
PO5	3	Effectively utilized modern engineering tools including Arduino IDE, LabVIEW, web technologies, and embedded hardware platforms for system implementation.
PO6	2	Considered efficient liquid utilization and automated process control techniques relevant to sustainable industrial automation practices.
PO7	2	Followed professional engineering practices and ethical implementation during system design, testing, and documentation.
PO8	3	Worked collaboratively as a multidisciplinary project team during circuit design, coding, hardware integration, and report preparation.
PO9	3	Demonstrated effective technical communication through project presentations, documentation, dashboard interface development, and system demonstrations.
PO10	2	Managed project planning, component selection, budgeting, and implementation activities within the available project timeline and resources.
PO11	3	Demonstrated self-learning and adaptability by exploring IoT communication, PWM-based dosing control, embedded web servers, and sensor calibration techniques.

Table 7.1: Mapping of Project to Program Outcomes

ANNEXURE B: MAPPING TO PROGRAM SPECIFIC OUTCOMES

LIST OF PROGRAM SPECIFIC OUTCOMES

- PSO1: Ability to design electronic instrumentation systems
- PSO2: Ability to implement control systems
- PSO3: Ability to acquire interdisciplinary technical skills

MAPPING THE PROJECT TO PROGRAM SPECIFIC OUTCOMES

PSO	Correlation	Justification
PSO1	3	Developed proficiency in building embedded instrumentation systems using ESP32, load cell sensors, motor drivers, and automation hardware.
PSO2	3	Applied control system optimization techniques including PWM-based pump control and automated dosing logic for industrial process applications.
PSO3	3	Enhanced practical skills in embedded programming, IoT integration, hardware prototyping, sensor interfacing, and automation system development.

Table 7.2: Mapping of Project to Program Specific Outcomes

ANNEXURE C: MAPPING TO SUSTAINABLE DEVELOPMENT GOALS

LIST OF SUSTAINABLE DEVELOPMENT GOALS (SDGS)

- SDG 1: No Poverty
- SDG 2: Zero Hunger
- SDG 3: Good Health and Well-being
- SDG 4: Quality Education
- SDG 5: Gender Equality
- SDG 6: Clean Water and Sanitation
- SDG 7: Affordable and Clean Energy
- SDG 8: Decent Work and Economic Growth
- SDG 9: Industry, Innovation and Infrastructure
- SDG 10: Reduced Inequalities
- SDG 11: Sustainable Cities and Communities
- SDG 12: Responsible Consumption and Production
- SDG 13: Climate Action
- SDG 14: Life Below Water
- SDG 15: Life on Land
- SDG 16: Peace, Justice and Strong Institutions
- SDG 17: Partnerships for the Goals

MAPPING THE PROJECT TO SUSTAINABLE DEVELOPMENT GOALS

SDG	Relevance to Project
SDG 4: Quality Education	The project promotes practical learning in embedded systems, industrial automation, IoT communication, sensor interfacing, and process control engineering through hands-on implementation and experimentation.
SDG 7: Affordable and Clean Energy	The developed prototype utilizes low-power embedded hardware and compact DC pumps, contributing towards energy-efficient automation system development.
SDG 8: Decent Work and Economic Growth	The system supports industrial automation practices that can improve operational efficiency, reduce manual intervention, and enhance productivity in small-scale process industries.
SDG 9: Industry, Innovation and Infrastructure	The project promotes industrial innovation through the implementation of embedded automation, IoT-based monitoring, intelligent liquid dosing, and web-based control techniques.
SDG 12: Responsible Consumption and Production	The automated weight-based dosing mechanism minimizes liquid wastage and improves process accuracy, thereby supporting efficient resource utilization and responsible production practices.
SDG 13: Climate Action	The reduction of excess material usage and process wastage through controlled automation contributes indirectly towards sustainable industrial operation and environmental conservation.

Table 7.3: Mapping of Project to Sustainable Development Goals